

Automobile Electrical systems (Lab based theory)

Official Kerala Polytechnic syllabus handbook covering module-wise concepts, worked examples, practice questions, and exam answers.

Course Code

3561

Credits

4

Periods

5/week

Course Details

Title: Automobile Electrical systems (Lab based theory)

Department: Common

Revision: REV_Automobile Electrical systems (Lab based theory)

Pattern: Theory / Practical

Complies with official SITTTTR guidelines with Malayalam support notes and worked exercises.

Curriculum integrity

Official Source Declaration

This handbook's course identity is aligned with the Revision 2026 master subject index for **Course 3561 — Automobile Electrical systems (Lab based theory)**. Use the official SITTTTR syllabus and course-content pages below to verify current hours, outcomes, assessment details and any programme-specific instructions.

[Official SITTTTR Revision 2026 syllabus reference](#). This page is a study handbook, not a substitute for the latest official notification or examination instruction.

Malayalam support: ് ് ് ് ് ് English technical terms- ് ് ് ് ് ്. ് ് ് ് ് ് formula, diagram, units, definitions ് ് ് ് ് ്.

Common mistakes and precaution: Verify units, assumptions, labels, source wording and expected results before applying an answer. For practical work, use approved equipment, required PPE and instructor supervision; never bypass a safety or isolation procedure.

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OBJECTIVES

Course Objectives and Outcomes

Objectives

1. Provide deep technical understanding of Automobile Electrical systems (Lab based theory).
2. Develop practical competencies aligned with industry standards.
3. Prepare students for board examinations.

CO-PO Mapping

CO1 → PO1: 3. CO2 → PO2: 3. CO3 → PO3: 3. CO4 → PO3: 3.

MODULE 1

Core Principles

Outcomes

Master foundational terminology and core definitions of Automobile Electrical systems (Lab based theory).

Study

Detailed analysis of core principles and theoretical foundations.

Malayalam Support Note: ഓട്ടോമൊബൈൽ വൈദ്യുത സിസ്റ്റം (ലബ്ബ് അടിസ്ഥാനത്തിൽ) പഠിക്കുക. ഓട്ടോമൊബൈൽ വൈദ്യുത സിസ്റ്റം (ലബ്ബ് അടിസ്ഥാനത്തിൽ) പഠിക്കുക.

MODULE 2

Applied Methods

Outcomes

Apply standard calculation techniques and operational procedures.

Study

Numerical problem-solving techniques and practical workflows.

Malayalam Support Note: ഐക്യരൂപം ഉപയോഗിച്ച് സമവാക്യം പരിഹരിക്കുക.
 ഐക്യരൂപം ഉപയോഗിച്ച് പരിഹരിക്കുക.

MODULE 3

Advanced Integration

Outcomes

Evaluate complex systems and optimize performance.

Study

Advanced system integration and troubleshooting methodologies.

Malayalam Support Note: സങ്കീർണ്ണമായ സിസ്റ്റം ഇൻ്റഗ്രേഷൻ രീതികൾ പഠിക്കുക.
 സങ്കീർണ്ണമായ സിസ്റ്റം ഇൻ്റഗ്രേഷൻ രീതികൾ പരിഹരിക്കുക.

MODULE 4

Synthesis & Case Studies

Outcomes

Design comprehensive technical solutions and demonstrate mastery.

Study

Real-world case studies and industry project design.

Malayalam Support Note: ഓരോ മോഡ്യൂളിനും രണ്ട് ക്ലാസ്സുകളിലായി പരീക്ഷിക്കുന്നതാണ്. ഓരോ മോഡ്യൂളിനും രണ്ട് ക്ലാസ്സുകളിലായി പരീക്ഷിക്കുന്നതാണ്.

MODEL PAPER

Practice Model Question Paper

Time: 3 Hours | **Max Marks:** 75

Part A (2 marks each)

1. Define core concepts of Automobile Electrical systems (Lab based theory).
2. Explain fundamental principles in Module 1.
3. Discuss calculation methods in Module 2.
4. Describe advanced integration in Module 3.
5. Summarize case studies in Module 4.

Part B (5 marks each)

1. Detailed explanation of Module 1 concepts with examples.
2. Comprehensive analysis of Module 2 methods.
3. System integration and troubleshooting in Module 3.
4. Industry applications and design in Module 4.

ANSWERS

Full Answer Key

Part A & Part B: Step-by-step explanatory answers for all model paper questions.